HOMEWORK -3

1.Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.

CODE:

```
#include <stdio.h>
#include <stdlib.h>

int main(){
   int n;

   printf("Enter number of elements: ");
   scanf("%d",&n);

   int *arr=(int*)malloc(n*sizeof(int));

   printf("Enter the elements:\n");
   for(int i=0; i<n; i++){
      scanf("%d",&arr[i]);
   }

   int max = arr[0];
```

```
    for(int i=0; i<n; i++){
        if(arr[i]>max){
            max= arr[i];
        }
    }

    printf("Largest element: %d",max);

    free(arr);

    return 0;
}
```

OUTPUT:

```
Enter number of elements: 5
Enter the elements:
56
78
23
45
12
Largest element: 78
```

2.Print an array in reverse order using pointers.

CODE:

```
#include <stdio.h>

int main(){
   int n, arr[100];

   printf("Enter size of array: ");
   scanf("%d",&n);
   printf("Enter the elements: ");

   for(int i=0; i<n; i++){
      scanf("%d",&arr[i]);
   }

   int *ptr = arr;

   printf("Reversed array: ");

   for(int i=n-1; i>=0; i--){
      printf("%d ",*(ptr+i));
   }
```

```
    return 0;
}
```

OUTPUT:

```
Enter size of array: 5
Enter the elements: 45
56
23
78
12
Reversed array: 12 78 23 56 45
```

3.Write a C program to store the information of students using structure and sort them using name.

CODE:

```
#include <stdio.h>
#include <string.h>

struct Students{
    int reg_no;
    char name[30];
    float cgpa;
};
```

```
int main(){
    int n;

    struct Students s[100],temp;

    printf("Enter number of students: ");
    scanf("%d",&n);

    printf("Enter Details:\n");

    for(int i=0; i<n; i++){
        printf("Roll number: ");
        scanf("%d",&s[i].reg_no);

        printf("Name: ");
        scanf("%s",s[i].name);

        printf("CGPA: ");
        scanf("%f",&s[i].cgpa);
    }

    for(int i=0; i<n-1; i++){
```

```
    for(int j=i+1; j<n; j++){
        if(strcmp(s[i].name, s[j].name)>0){
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }

  }

  printf("After sorting based on name: \n");
  for(int i=0; i<n; i++){
    printf("\nRoll no: %d\nName: %s\nCGPA:
%.2f",s[i].reg_no,s[i].name,s[i].cgpa);
  }

  return 0;
}
```

OUTPUT:

```
Enter number of students: 2
Enter Details:
Roll number: 5028
Name: Sangavi
```

CGPA: 9.13

Roll number: 5116

Name: Pavatharani

CGPA: 9.12

After sorting based on name:

Roll no: 5116

Name: Pavatharani

CGPA: 9.12

Roll no: 5028

Name: Sangavi

CGPA: 9.13

4. Write a C program to add n distances given in inch-feet system using structures.

CODE:

```
#include <stdio.h>

struct Distance {
   int feet;
   int inch;
};
```

```
int main() {
    int n;
    struct Distance d[100], sum = {0, 0};

    printf("Enter number of distances: ");
    scanf("%d", &n);

    printf("Enter distances (feet inch):\n");

    for (int i = 0; i < n; i++) {
        printf("Distance %d: ", i + 1);
        scanf("%d %d", &d[i].feet, &d[i].inch);

        sum.feet += d[i].feet;
        sum.inch += d[i].inch;
    }


    sum.feet += sum.inch / 12;
    sum.inch = sum.inch % 12;

    printf("\nTotal Distance = %d feet %d inches\n",
           sum.feet, sum.inch);
```

```
    return 0;
}
```

OUTPUT:

```
Enter number of distances: 3
Enter distances (feet inch):
Distance 1: 5 8
Distance 2: 3 7
Distance 3: 2 10

Total Distance = 12 feet 1 inches
```

5.Write a C program to add two complex numbers by passing structure to a function.

CODE:

```
#include <stdio.h>

struct Complex{
    int real;
    int img;
};

void add(struct Complex n1, struct Complex n2){
```

```
    struct Complex result;
    result.real = n1.real + n2.real;
    result.img = n1.img + n2.img;

    printf("Result =%d + %di",result.real,result.img);
}

int main(){
   struct Complex num1, num2;

    printf("Enter complex number1 (real, imag): ");
    scanf("%d %d",&num1.real, &num1.img);

    printf("Enter complex number2 (real, imag): ");
    scanf("%d %d",&num2.real, &num2.img);

    add(num1, num2);
}
```

OUTPUT:

```
Enter complex number1 (real, imag): 54 8
Enter complex number2 (real, imag): 32 6
Result =86 + 14i
```

6. Write a C program to maintain inventory management system using structures.

CODE:

```
#include <stdio.h>

struct product{
   int id;
   char name[30];
   int quantity;
   float price;
};

int main(){
   struct product p[100];
   int n;
   float sum=0;

   printf("Enter number of products: ");
   scanf("%d", &n);

   printf("Enter details:\n");

   for(int i=0; i<n; i++){
```

```
        printf("Product: %d",i+1);

        printf("\nProd_id: ");
        scanf("%d",&p[i].id);

        printf("Prod_name: ");
        scanf("%s",p[i].name);

        printf("Qunantity: ");
        scanf("%d",&p[i].quantity);

        printf("Price: ");
        scanf("%f",&p[i].price);
    }

    for(int i=0; i<n; i++){
         sum += (p[i].quantity*p[i].price);
    }

    printf("\nTotal price: %.2f",sum);

    return 0;
}
```

OUTPUT:

Enter number of products: 2

Enter details:

Product: 1

Prod_id: 1234

Prod_name: A4sheet

Qunantity: 1

Price: 999

Product: 2

Prod_id: 7890

Prod_name: trolley

Qunantity: 3

Price: 3999

Total price: 12996.00

7. Write a C program for decimal to binary, octal, hexadecimal conversion.

CODE:

```
#include <stdio.h>

int main(){
   int n;
```

```
int rem1=0, rem2=0, rem3[32];

printf("Enter the decimal number: ");
scanf("%d",&n);

int temp = n;

while(temp!=0){
   rem1 = rem1*10 +(temp%2);
   temp = temp/2;
}

int binary=0;
while(rem1!=0){
   binary = binary*10 +(rem1%10);
   rem1 = rem1/10;

}

temp =n;

while(temp!=0){
   rem2 = rem2*10 +(temp%8);
```

```
    temp = temp/8;
}

int octal=0;
while(rem2!=0){
    octal = octal*10 +(rem2%10);
    rem2 = rem2/10;

}

temp =n;
int i=0;

    while (temp != 0) {
    int x = temp % 16;

    if (x < 10)
        rem3[i] = x + '0';
    else
        rem3[i] = x - 10 + 'A';

    i++;
    temp = temp / 16;
```

```
    }

    rem3[i] = '\0';

    printf("Binary number:%d",binary);
    printf("\nOctal number:%d",octal);
    printf("\nHexadecimal number: ");

    for(i=i-1; i>=0; i--){
        printf("%c",rem3[i]);
    }

    return 0;

}
```

OUTPUT:

```
Enter the decimal number: 255
Binary number:11111111
Octal number:377
Hexadecimal number: FF
```

8. Write a C program to calculate the diference between two time periods.

CODE:

```
#include <stdio.h>

struct time{
   int hr;
   int min;
   int sec;
};

int main(){
   struct time t1, t2, t3;
   int diff;

   printf("Enter Time period 1 (hr,min,sec): ");
   scanf("%d %d %d",&t1.hr,&t1.min,&t1.sec);

   printf("Enter Time period 2 (hr,min,sec): ");
   scanf("%d %d %d",&t2.hr,&t2.min,&t2.sec);

   t1.min += t1.hr*60;
   t1.sec += t1.min*60;

   t2.min += t2.hr*60;
```

```
   t2.sec += t2.min*60;



   if(t1.sec > t2.sec){
     diff = t1.sec - t2.sec;
   }
   else{
      diff = t2.sec - t1.sec;
   }


   t3.hr = diff/3600;
   int rem = diff%3600;


   t3.min = rem/60;
   t3.sec = rem%60;


   printf("Difference between time periods: %d hours %d min %d
sec",t3.hr,t3.min,t3.sec);


   return 0;


}
```

OUTPUT:

Enter Time period 1 (hr,min,sec): 2 42 52

Enter Time period 2 (hr,min,sec): 5 67 23

Difference between time periods: 3 hours 24 min 31 sec

9. Write a C program to implement employee management system using structures.

CODE:

```
#include <stdio.h>

struct employees{
    int emp_ID;
    char name[10];
    float salary;
};

int main(){
   int n;
   float hra, da,total,high =-1;
   char high_emp[10];

   struct employees s[100];
   struct employees *ptr;
```

```
printf("Enter number of employees working:");
scanf("%d",&n);
for(int i = 0; i < n; i++){
   ptr =&s[i];

   printf("Enter ID:");
   scanf("%d",&ptr->emp_ID);

   printf("Enter name:");
   scanf("%s",ptr->name);

   printf("Enter basic salary:");
   scanf("%f",&ptr->salary);
}

for(int i = 0; i < n; i++){
   ptr =&s[i];
   printf("\nEmployee ID: %d",ptr->emp_ID);
   printf("\nName: %s",ptr->name);
   printf("\nSalary: %.2f",ptr->salary);

   ptr->salary = ptr->salary * 1.10;
   hra = 0.20*ptr->salary;
```

```
      da = 0.10*ptr->salary;
      total = ptr->salary+ hra + da;

      printf("\nHRA amount:%.2f\nDA amount:%.2f\nGross
salary:%.2f",hra,da,total);

      if(total > high){
       high = total;
       int j =0;

       while(ptr->name[j] !='\0'){
         high_emp[j] =ptr->name[j];
         j++;
       }

       high_emp[j]='\0';
      }
       printf("\n");

   }
   printf("\n\nEmployee with highest salary:%s\nHighest
salary:%.2f",high_emp,high);
```

```
    return 0;
}
```

OUTPUT:

```
Enter number of employees working:2
Enter ID:28
Enter name:Sangavi
Enter basic salary:80000
Enter ID:116
Enter name:Pava
Enter basic salary:72000

Employee ID: 28
Name: Sangavi
Salary: 80000.00
HRA amount:17600.00
DA amount:8800.00
Gross salary:114400.00

Employee ID: 116
Name: Pava
Salary: 72000.00
HRA amount:15840.00
DA amount:7920.00
```

Gross salary:102960.00

Employee with highest salary:Sangavi

Highest salary:114400.00

10. Write a C program to find the area and perimeter of a rectangle, triangle using function pointers.

CODE:

```
#include <stdio.h>

struct Rectangle {
   float length;
   float breadth;
};

struct Triangle {
   float base;
   float height;
   float side1;
   float side2;
   float side3;
};
```

```
void fun1(struct Rectangle r){
   float per = 2*(r.length + r.breadth);
   float area = r.length * r.breadth;

   printf("\nRectangle: ");

   printf("\nArea = %.2f\n", area);
   printf("Perimeter = %.2f\n", per);
}

void fun2(struct Triangle t){
   float per = 0.5*(t.base+ t.height);
   float area = t.side1 * t.side2 * t.side3;

    printf("\nTriangle: ");

   printf("\nArea = %.2f\n", area);
   printf("Perimeter = %.2f\n", per);
}


int main(){
    struct Rectangle r1;
```

```
 struct Triangle t1;

printf("Rectangle: ");

printf("\nEnter length: ");
scanf("%f",&r1.length);

printf("Enter breadth: ");
scanf("%f",&r1.breadth);

printf("\nTriangle: ");

printf("\nEnter base : ");
scanf("%f",&t1.base);

printf("Enter height : ");
scanf("%f",&t1.height);

printf("Enter side 1 : ");
scanf("%f",&t1.side1);

printf("Enter side 2 : ");
scanf("%f",&t1.side2);
```

```
    printf("Enter side 3 : ");
    scanf("%f",&t1.side3);

    void (*ptr1)(struct Rectangle);
    ptr1 = fun1;
    void (*ptr2)(struct Triangle);
    ptr2 = fun2;

    ptr1(r1);
    ptr2(t1);

    return 0;

}
```

OUTPUT:

```
Rectangle:
Enter length: 12.5
Enter breadth: 5.9

Triangle:
Enter base : 7.9
Enter height : 9.0
```

Enter side 1 : 3.2

Enter side 2 : 5.4

Enter side 3 : 1.2

Rectangle:

Area = 73.75

Perimeter = 36.80

Triangle:

Area = 20.74

Perimeter = 8.45